The Gram stain

THE GRAM STAIN (sample notes)

Steps

1. Crystal violet (primary stain) is applied to all cells.
2. Gram's iodine is applied as a mordant, forming a crystal violet-iodine complex.
3. Decolouriser (ethanol or acetone) is applied briefly.
4. Safranin (counterstain) is applied last.

Result

- Gram-positive cells retain the crystal violet complex (thick peptidoglycan traps it) and appear purple.
- Gram-negative cells lose the complex in the decolouriser (the outer membrane is disrupted and the thin wall cannot retain it) and take up safranin, appearing pink or red.

Common mistakes

- Over-decolourising a Gram-positive culture can make it falsely appear Gram-negative.
- Old cultures can stain Gram-variable.

Study tip: remember "purple = peptidoglycan thick, pink = outer membrane".